

Multi-Factor Authentication (MFA)

Task 1

Enable Multi-Factor Authentication (MFA) on your personal AWS or Azure cloud account using the built-in authenticator app option, and take a screenshot of the MFA setup confirmation page.

Steps followed:

1. Logged in to the AWS Management Console using my root/IAM account and went to “Security credentials” from the account name menu in the top-right corner.
2. In the “Multi-factor authentication (MFA)” section, clicked “Assign MFA device”.
3. Selected “Authenticator app” as the MFA device type and clicked “Next”.
4. Opened an authenticator app (Google Authenticator) on my phone and scanned the QR code shown by AWS to link the account.
5. Entered two consecutive 6-digit codes generated by the authenticator app into the “MFA code 1” and “MFA code 2” fields to verify the setup, then clicked “Add MFA”.
6. AWS displayed a confirmation message that the MFA device was successfully assigned, and the Security credentials page now showed the authenticator app listed as an active MFA device.

(Screenshot: MFA setup confirmation page showing the authenticator app successfully assigned — attached separately as evidence.)

Task 2

Simulate a login to your cloud account from a new browser or incognito mode and document the steps and prompts you receive with MFA enabled.

Steps followed:

1. Opened a new incognito/private browser window to simulate a fresh login session with no saved cookies or session data.
2. Went to the AWS sign-in page and entered the account ID/alias, IAM username, and password as usual.
3. After the password was accepted, AWS did not log me in directly — instead it showed an additional prompt asking for the “MFA code” from my registered authenticator app.
4. Opened the authenticator app, read the current 6-digit time-based code, and entered it into the MFA prompt within the 30-second validity window.
5. Only after the correct MFA code was submitted did AWS grant access and load the Management Console dashboard, confirming that password alone was not sufficient to log in.

(Screenshot: Incognito login flow showing the MFA code prompt after password entry — attached separately as evidence.)

Task 3

Disable MFA temporarily on your cloud account, attempt to log in again, and note the difference in the login experience and security prompts. Hint: Compare the number of steps and any warnings shown with and without MFA.

Steps followed:

1. Went back to “Security credentials”, found the assigned authenticator device under the MFA section, and clicked “Remove” to temporarily deactivate MFA on the account.
2. Logged out completely and attempted to log back in from the same incognito browser window.
3. This time, after entering the username and password, AWS logged me straight into the console — no second code prompt appeared at all.
4. Noted that the login process was noticeably shorter (just one step — password only), and the account overview page also showed a security recommendation/warning banner suggesting that MFA should be enabled to better protect the account.

Comparison observed:

1. With MFA enabled — 2 steps to log in (password + authenticator code), no security warning shown, and access denied if the correct time-based code was not provided.
2. With MFA disabled — 1 step to log in (password only), and AWS displayed a visible “Root user does not have MFA enabled” style warning encouraging MFA setup, since a single leaked password would be enough for anyone to gain full account access.

Task 4

Write a short scenario (3-5 lines) describing how an attacker might try to access a cloud account without MFA, and explain how enabling MFA would block or alert you to this attempt.

Answer:

Suppose an attacker obtains a user's cloud account password through a phishing email, a data breach on another site where the same password was reused, or malware that logs keystrokes. Without MFA enabled, this password alone is enough for the attacker to log straight into the AWS or Azure console and gain full access to resources, data, and billing settings, often without the real owner noticing anything unusual. With MFA enabled, even after entering the correct password, the attacker would be blocked at the second step because they do not have the victim's physical authenticator app to generate the required time-based one-time code. Many cloud providers also send a login notification or flag the failed second-factor attempt as suspicious activity, alerting the real account owner that someone tried to sign in with their password so they can change it immediately. In this way, MFA turns a single stolen password from a complete account takeover into just a failed, and traceable, login attempt.